

# Stability Theorem for Causal SISO LTI Systems

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# Definition of Bounded Input/Output

A causal SISO LTI system's input  $u(t)$  is **bounded** if

$$\|u\|_{\infty} := \sup_{t \in [0, \infty)} |u(t)| < \infty$$

Similarly, its output  $y(t)$  is **bounded** if

$$\|y\|_{\infty} := \sup_{t \in [0, \infty)} |y(t)| < \infty$$

## Examples

$u(t) = t$  is not bounded because  $\|u\|_{\infty} = \sup_{t \in [0, \infty)} |t| = \infty$

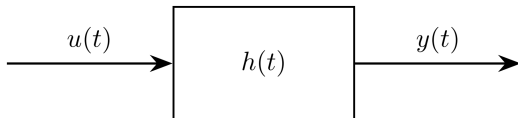
$u(t) = \sin t$  is bounded because  $\|u\|_{\infty} = \sup_{t \in [0, \infty)} |\sin t| = 1 < \infty$

$u(t) = \frac{t}{1+t}$  is bounded because  $\|u\|_{\infty} = \sup_{t \in [0, \infty)} \left| \frac{t}{1+t} \right| = 1 < \infty$  (maximum does not exist)

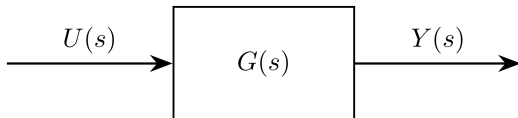
# Definition of Stability

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Consider a causal SISO LTI system shown in  $t$  domain:



or equivalently (recall  $G(s) := \mathcal{L}[h(t)]$ ) in  $s$  domain:



We say this system is bounded input bounded output (BIBO) stable, or internally stable, or stable, if

bounded input implies bounded output, i.e.,  $\|u\|_{\infty} < \infty \Rightarrow \|y\|_{\infty} < \infty$

## Stability Theorem: Statement

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For a causal SISO LTI system with impulse response function  $h(t)$  and rational transfer function  $G(s)$ , the following statements are equivalent

(i) The system is BIBO stable or internally stable or stable

(ii) The impulse response function is absolutely integrable, i.e.,

$$\|h\|_1 := \int_0^{\infty} |h(t)| dt < \infty$$

(iii) All poles of  $G(s)$  are in open left half plane (LHP)  $\mathbb{C}_-$

## Stability Theorem: Proof: (ii) $\Rightarrow$ (i)

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Since  $y(t) = h(t) * u(t) = \int_0^\infty h(\tau)u(t - \tau)d\tau$  (see Lec. 6 p. 1-7), therefore

$$\begin{aligned} |y(t)| &= \left| \int_0^\infty h(\tau)u(t - \tau)d\tau \right| \\ &\leq \int_0^\infty |h(\tau)u(t - \tau)| d\tau \quad (\text{applying triangle inequality}) \\ &= \int_0^\infty |h(\tau)| |u(t - \tau)| d\tau \\ &\leq \|u\|_\infty \int_0^\infty |h(\tau)| d\tau \end{aligned}$$

So  $\|h\|_1 < \infty \Rightarrow \|y\|_\infty < \infty$ , showing (ii)  $\Rightarrow$  (i)

## Stability Theorem: Proof: (i) $\Rightarrow$ (ii)

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Contraposition: we will show  $\int_0^\infty |h(t)|dt = \infty \Rightarrow \text{not BIBO}$

Consider specific  $u(t) := -\text{sign}(h(-t))$ . Clearly  $\|u\|_\infty = 1 < \infty$ , i.e.,  $u(t)$  is bounded

However,  $y(0) = \int_0^\infty h(0 - \tau)u(\tau)d\tau = \int_0^\infty |h(\tau)|d\tau = \infty$ , i.e.,  $y(t)$  is not bounded

This shows (i)  $\Rightarrow$  (ii)

## Stability Theorem: Proof for (iii) $\Rightarrow$ (ii)

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Because the system is causal SISO LTI, the transfer function  $G(s)$  is **proper**

Suppose that  $G(s)$  has  $\nu$  **distinct** poles:  $p_1, p_2, \dots, p_\nu \in \mathbb{C}$  with multiplicities  $n_1, n_2, \dots, n_\nu$  such that  $n_1 + n_2 + \dots + n_\nu = n$

Then, by partial fraction expansion, we can write  $G(s)$  in residue form (Lec. 8 p. 4)

$$G(s) = k + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} \frac{r_{i\ell}}{(s - p_i)^\ell}$$

where the residues  $r_{i\ell} \in \mathbb{C}$ , and the **real** constant  $k := \lim_{s \rightarrow \infty} G(s) < \infty$ . Clearly,  $k = 0$  if  $G$  is strictly proper, and  $\neq 0$  otherwise

## Stability Theorem: Proof for (iii) $\Rightarrow$ (ii) continued

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Taking  $\mathcal{L}^{-1}[\cdot]$  to both sides of the residue form, we get

$$\begin{aligned} h(t) &= \mathcal{L}^{-1}[G(s)] = k\mathcal{L}^{-1}[1] + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} r_{i\ell} \mathcal{L}^{-1} \left[ \frac{1}{(s - p_i)^\ell} \right] \\ &= k\delta(t) + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} r_{i\ell} e^{p_i t} \frac{t^{\ell-1}}{(\ell-1)!} \end{aligned}$$

Poles in real-imaginary part form:  $p_i = \sigma_i + j\omega_i$  with  $\sigma_i, \omega_i \in \mathbb{R}$  for all  $i = 1, 2, \dots, n$

Since  $|e^{p_i t}| = e^{\sigma_i t}$ , by triangle inequality and that  $|\delta(t)| = \delta(t)$  (Lec. 5 p. 6), we have

$$|h(t)| \leq |k|\delta(t) + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} |r_{i\ell}| e^{\sigma_i t} \frac{t^{\ell-1}}{(\ell-1)!}$$

## Stability Theorem: Proof for (iii) $\Rightarrow$ (ii) continued

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For  $p_i \in \mathbb{C}_- \Leftrightarrow \sigma_i < 0$ , we integrate both sides of the last inequality from  $t = 0$  to  $\infty$ :

$$\begin{aligned} \int_0^\infty |h(t)| \, dt &\leq |k| \underbrace{\int_0^\infty \delta(t) \, dt}_{=1 \text{ (Lec. 5 p. 6)}} + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} \frac{|r_{i\ell}|}{(\ell-1)!} \underbrace{\int_0^\infty t^{\ell-1} e^{\sigma_i t} \, dt}_{\substack{\text{Laplace transform of power} \\ = \frac{(\ell-1)!}{(-\sigma_i)^\ell}}} \\ &= |k| + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} \frac{|r_{i\ell}|}{(-\sigma_i)^\ell} < \infty \end{aligned}$$

This shows (iii)  $\Rightarrow$  (ii)

## Stability Theorem: Proof for (ii) $\Rightarrow$ (iii)

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Here, we are given  $\int_0^\infty |h(t)| dt < \infty$

For any  $s \in \overline{\mathbb{C}}_+$  (closed RHP where  $\text{Re}(s) \geq 0$ ), and for any  $t \in [0, \infty)$ ,

$$|e^{-st}| = e^{-\text{Re}(s)t} \leq 1$$

Therefore,  $|G(s)| = \left| \int_0^\infty h(t) e^{-st} dt \right| \leq \int_0^\infty |h(t)| dt < \infty$

This shows (ii)  $\Rightarrow$  (iii) since otherwise (i.e., if  $G$  had a pole  $p_i = \sigma_i + j\omega_i$  with  $\sigma_i \geq 0$ )  $|G(s)| \uparrow \infty$  as  $s \rightarrow p_i$ , thereby contradicting the bound

## Remarks

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- The stability theorem and its proof extend to MIMO LTI systems. In SISO case, slide 4 condition (ii) says  $\|h\|_1 < \infty$ . In MIMO case, arrange the impulse response integrals  $\int_0^\infty |h_{ij}(t)|dt$  in a matrix  $H$  of size  $p \times m$  (number of outputs  $\times$  number of inputs). Then condition (ii) becomes matrix 1-norm condition:  $\|H\|_1 < \infty$ . Condition (iii) generalizes: all poles of  $p \times m$  transfer matrix  $G(s)$  must lie in  $\mathbb{C}_-$ .
- In slide 3, the notions of BIBO stability and stability/internal stability coincide for LTI systems, but not for general LTV and nonlinear systems
- In slide 4 condition (iii), we tacitly assumed that  $G(s)$  has no pole-zero cancellations. If there are unstable pole-zero cancellations (e.g., terms like  $(s - 3)$ ) then the SISO LTI system is BIBO stable in the sense of (i)-(ii), but not internally stable in the sense of (iii). We then say: the SISO LTI realization is non-minimal